

Ex:No: 4A

Date:

Predicting Crop Disease Risk Using Decision Tree Algorithm

Problem Description:

An agricultural research institute monitors environmental conditions to reduce crop losses caused by plant diseases. Factors such as temperature, humidity, rainfall, and soil moisture strongly influence disease occurrence. The institute wants to **predict whether crops are at high or low disease risk** based on these conditions using a **Decision Tree model**, enabling early preventive measures.

Objective:

To build and evaluate a Decision Tree classification model in R to predict crop disease risk using environmental data with train-test split and accuracy calculation.

```
1  # Step 1: Create Dataset
2  crop_data <- data.frame(
3    Temperature = c(25,30,28,35,22,27,33,29,31,24,34,26),
4    Humidity = c(80,60,75,55,85,78,50,70,65,88,52,82),
5    Rainfall = c(120,40,90,30,150,100,20,85,60,140,25,110),
6    SoilMoisture = c("High","Medium","High","Low","High","High","Low","Medium","Medium","High","Low"
7    DiseaseRisk = c("High","Low","High","Low","High","High","Low","High","Low","High","Low","High")
8  )
9
10
11 cat("Crop Data:\n")
12 print(crop_data)
13
14 # Step 2: Convert to Factors
15 crop_data$SoilMoisture <- as.factor(crop_data$SoilMoisture)
16 crop_data$DiseaseRisk <- as.factor(crop_data$DiseaseRisk)
17
18 cat("\nStructure:\n")
19 str(crop_data)
20
21 # Step 3: Train-Test Split (70-30)
22 set.seed(123)
23 n <- nrow(crop_data)
24 train_index <- sample(1:n, 0.7 * n)
25
26 train_data <- crop_data[train_index, ]
27 test_data <- crop_data[-train_index, ]
28
29 cat("\nTraining rows:", nrow(train_data))
30 cat("\nTesting rows:", nrow(test_data), "\n")
31
32 # Step 4: Build Decision Tree
33 # install.packages("rpart")
34 library(rpart)
35
36 disease_tree <- rpart(DiseaseRisk ~ ., data = train_data, method = "class")
37
38 cat("\nDecision Tree Model:\n")
39 print(disease_tree)
40
41 # Step 5: Visualize Tree
42 # install.packages("rpart.plot")
43 library(rpart.plot)
44 rpart.plot(disease_tree)
45
46 # Step 6: Predict Disease Risk
47 predictions <- predict(disease_tree, test_data, type = "class")
48
49 cat("\nPredictions:\n")
50 print(predictions)
51 # Step 7: Confusion Matrix
52 conf_matrix <- table(Predicted = predictions, Actual = test_data$DiseaseRisk)
53
54 cat("\nConfusion Matrix:\n")
55 print(conf_matrix)
56
57 # Step 8: Accuracy Calculation
58 accuracy <- sum(diag(conf_matrix)) / sum(conf_matrix)
59
60 cat("\nAccuracy:", accuracy, "\n")
```

```
> source("~/active-rstudio-document")
Crop Data:
  Temperature Humidity Rainfall SoilMoisture DiseaseRisk
1           25         80      120         High         High
2           30         60       40         Medium        Low
3           28         75       90          High         High
4           35         55       30          Low          Low
5           22         85      150          High         High
6           27         78      100          High         High
7           33         50       20          Low          Low
8           29         70       85         Medium        High
9           31         65       60         Medium        Low
10          24         88      140          High         High
11          34         52       25          Low          Low
12          26         82      110          High         High

Structure:
'data.frame':  12 obs. of  5 variables:
 $ Temperature : num  25 30 28 35 22 27 33 29 31 24 ...
 $ Humidity     : num  80 60 75 55 85 78 50 70 65 88 ...
 $ Rainfall     : num  120 40 90 30 150 100 20 85 60 140 ...
 $ SoilMoisture: Factor w/ 3 levels "High","Low","Medium": 1 3 1 2 1 1 2 3 3 1 ...
 $ DiseaseRisk  : Factor w/ 2 levels "High","Low": 1 2 1 2 1 1 2 1 2 1 ...

Training rows: 8
Testing rows: 4

Decision Tree Model:
n= 8

node), split, n, loss, yval, (yprob)
* denotes terminal node

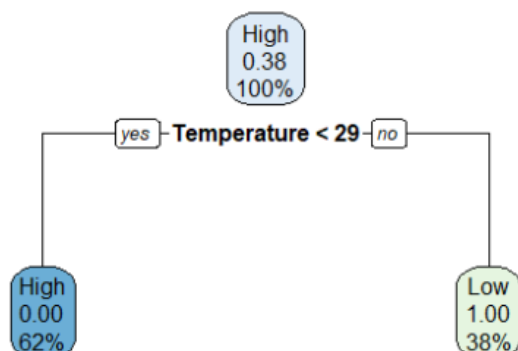
1) root 8 3 High (0.6250000 0.3750000) *

Predictions:
[1] High High High High
Levels: High Low
```

Confusion Matrix:

	Actual	
Predicted	High	Low
High	2	2
Low	0	0

Accuracy: 0.5



Ex:No: 4B

Date:

Decision Tree Algorithm Using Iris Dataset

Problem Description:

The Iris dataset contains measurements of iris flowers belonging to three species: *Setosa*, *Versicolor*, and *Virginica*. The goal is to use flower measurements such as sepal length, sepal width, petal length, and petal width to **classify the species of iris flowers** using a **Decision Tree algorithm**.

Objective:

To implement a Decision Tree classification model in R using the Iris dataset, perform train-test splitting, predict flower species, and evaluate model accuracy.

```
1 # Step 2: Load and Explore Dataset
2 data(iris)
3 print(head(iris))
4
5 set.seed(123)
6 library(caret)
7 train_index <- createDataPartition(iris$Species, p = 0.8, list = FALSE)
8 train_data <- iris[train_index, ]
9 test_data <- iris[-train_index, ]
10
11 cat("\nTraining rows:", nrow(train_data))
12 cat("\nTesting rows:", nrow(test_data), "\n")
13
14 # Step 4: Train Decision Tree Model
15 library(rpart)
16 model <- rpart(Species ~ ., data = train_data, method = "class")
17 cat("\nDecision Tree Model:\n")
18 print(model)
19
20 # Step 5: Visualize Decision Tree
21 library(rpart.plot)
22 rpart.plot(model)
23
24 # Step 6: Evaluate Model
25 predictions <- predict(model, test_data, type = "class")
26
27 cat("\nPredictions:\n")
28 print(predictions)
29
30 conf_matrix <- table(Predicted = predictions, Actual = test_data$Species)
31
32 cat("\nConfusion Matrix:\n")
33 print(conf_matrix)
34
35 accuracy <- sum(diag(conf_matrix)) / sum(conf_matrix)
36
37 cat("\nAccuracy:", accuracy, "\n")
```

First 6 rows of Iris Dataset:

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

Training rows: 120

Testing rows: 30

Decision Tree Model:

n= 120

node), split, n, loss, yval, (yprob)
* denotes terminal node

```

1) root 120 80 setosa (0.33333333 0.33333333 0.33333333)
 2) Petal.Length < 2.45 40 0 setosa (1.00000000 0.00000000 0.00000000) *
 3) Petal.Length >= 2.45 80 40 versicolor (0.00000000 0.50000000 0.50000000)
    6) Petal.Width < 1.75 42 3 versicolor (0.00000000 0.92857143 0.07142857) *
    7) Petal.Width >= 1.75 38 1 virginica (0.00000000 0.02631579 0.97368421) *
  
```

Predictions:

	1	2	6	16	23	34	35	38	44
setosa	47	51	60	64	73	74	87	91	94
versicolor	95	97	104	109	111	113	116	120	127
virginica	134	138	147						

Levels: setosa versicolor virginica

Confusion Matrix:

	Actual		
Predicted	setosa	versicolor	virginica
setosa	10	0	0
versicolor	0	10	2
virginica	0	0	8

Accuracy: 0.9333333

